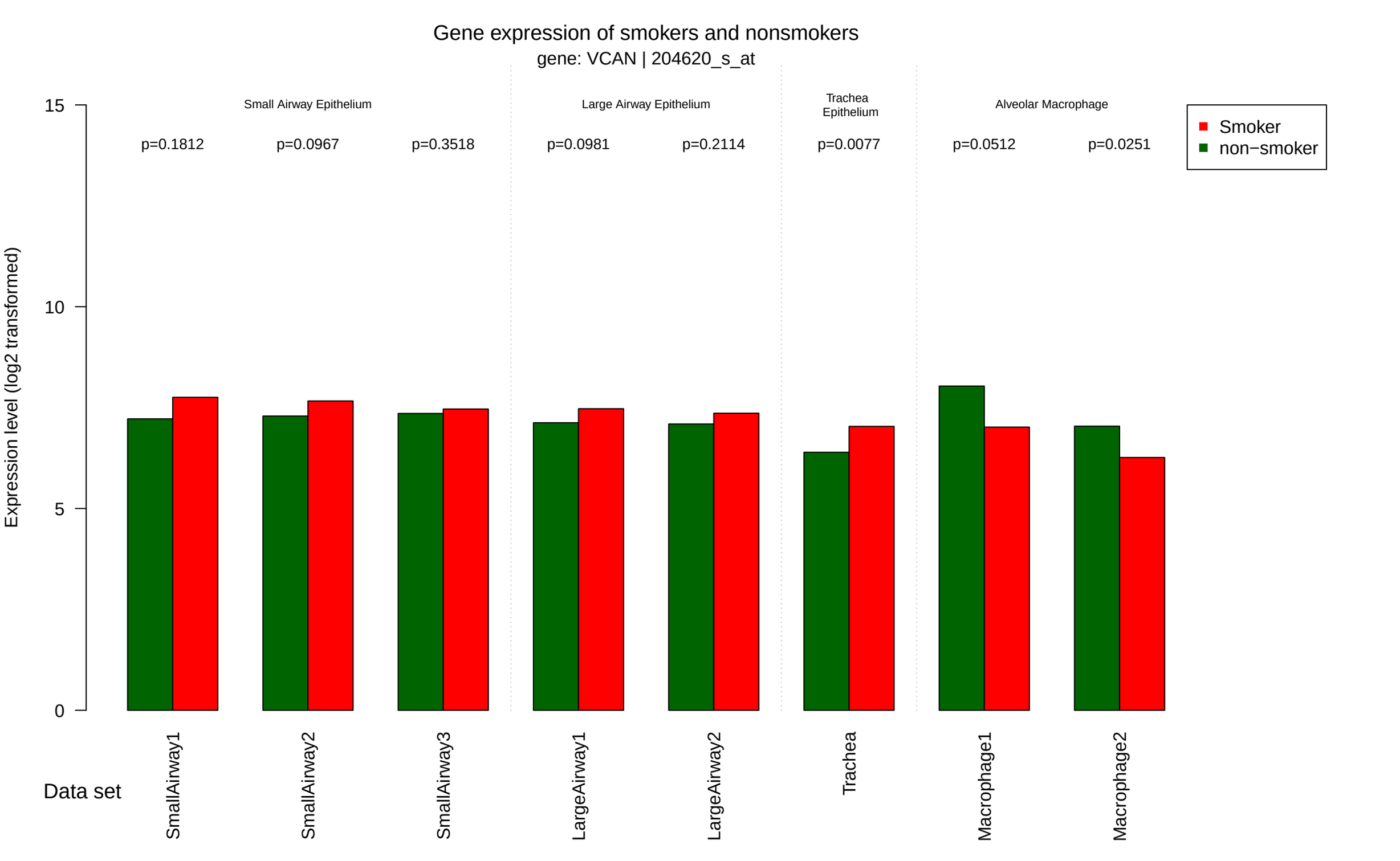
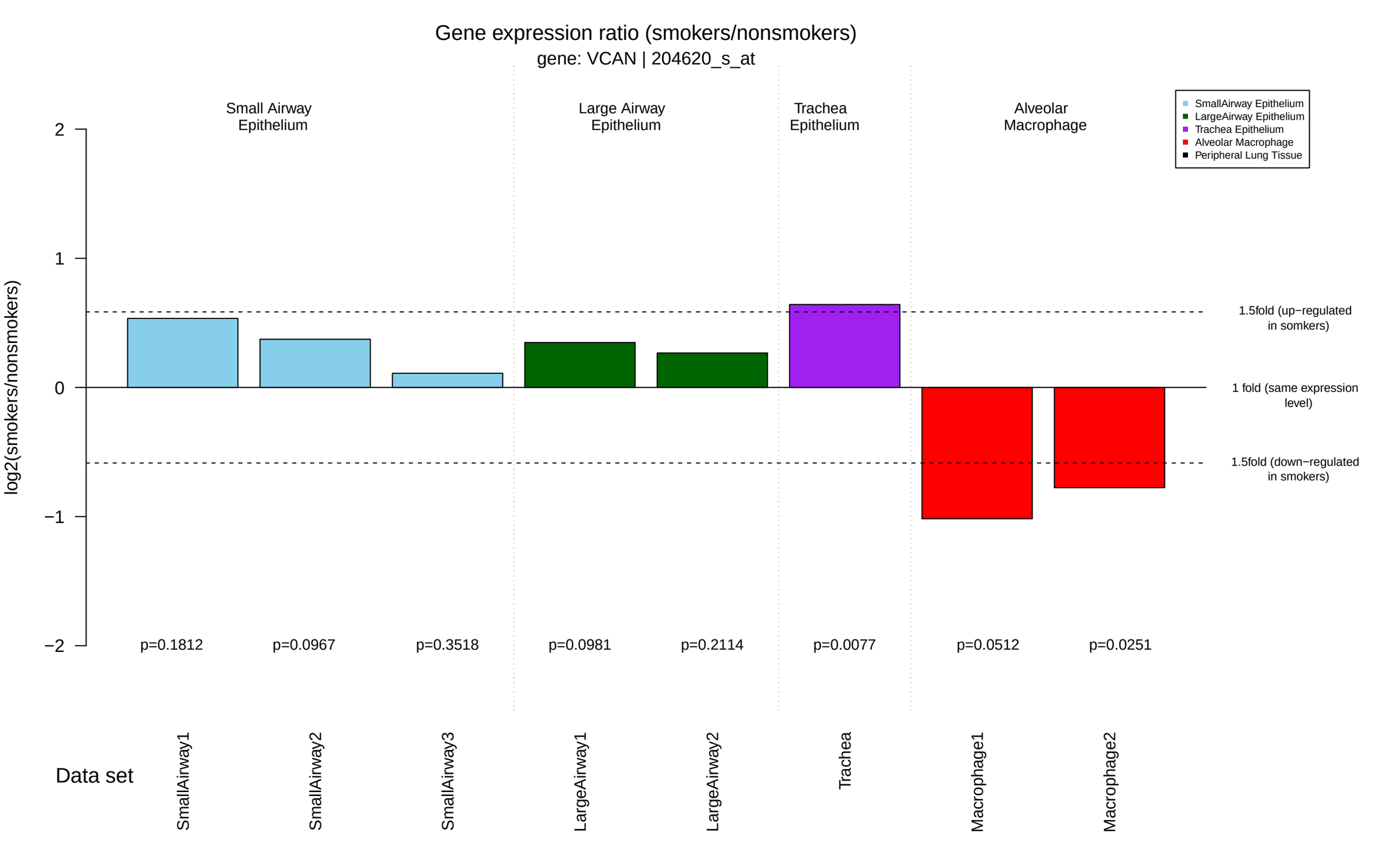
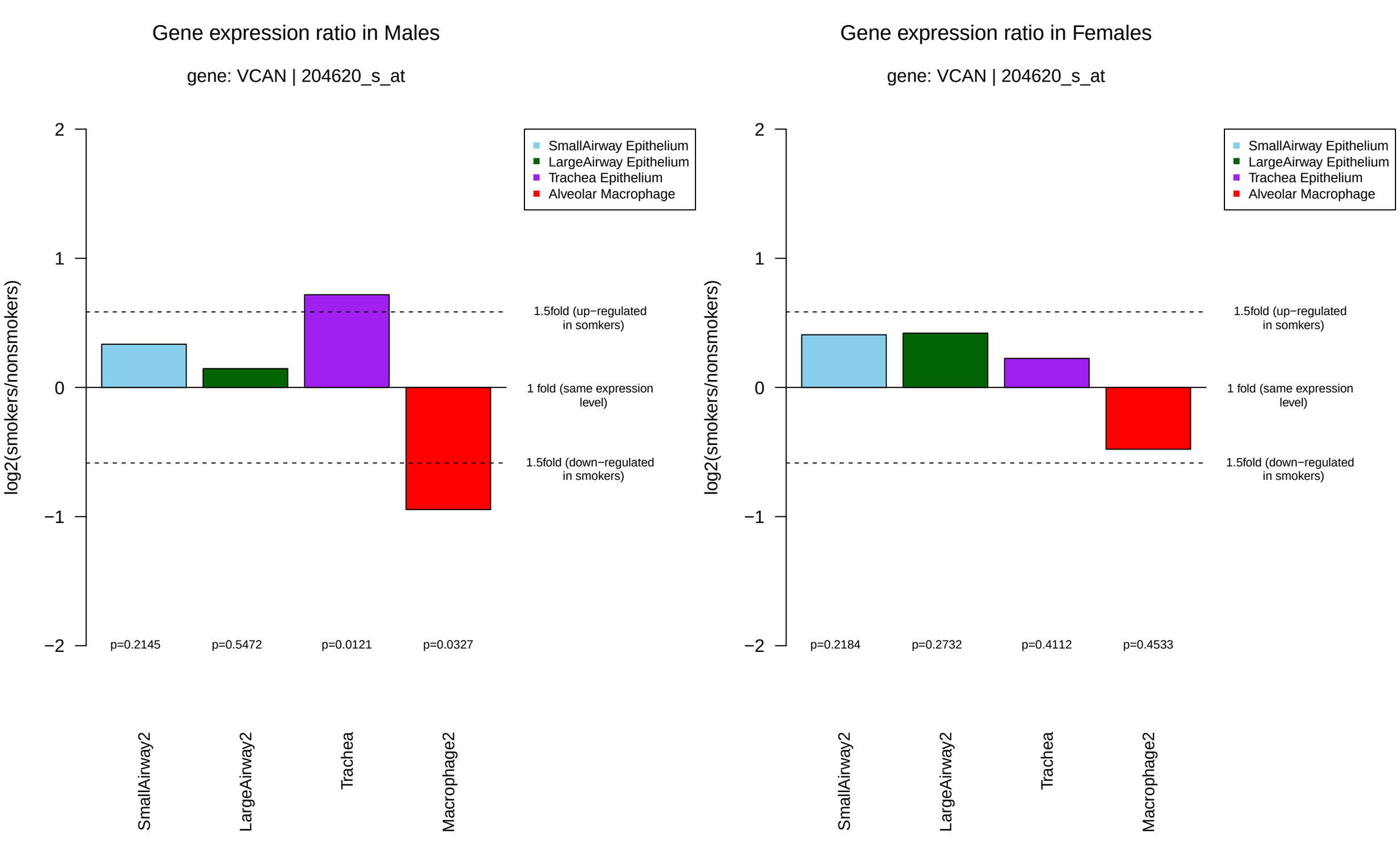
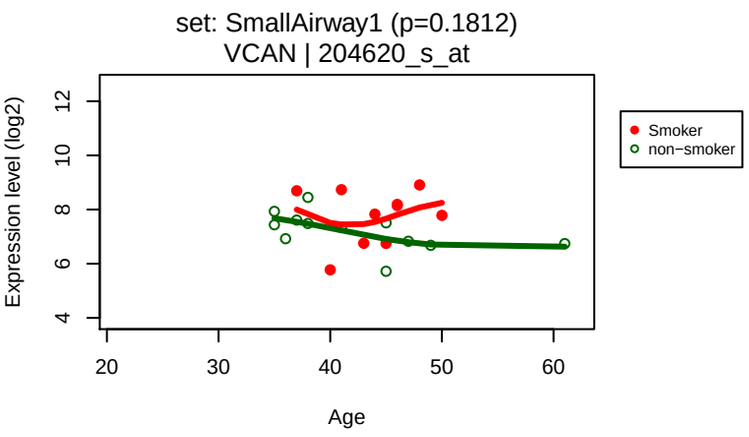
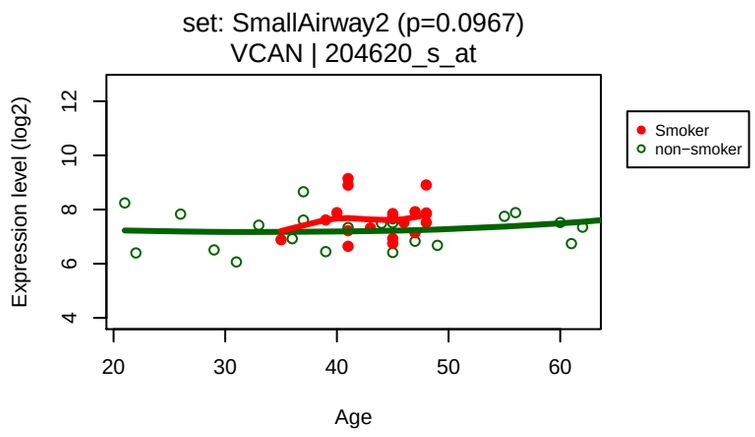
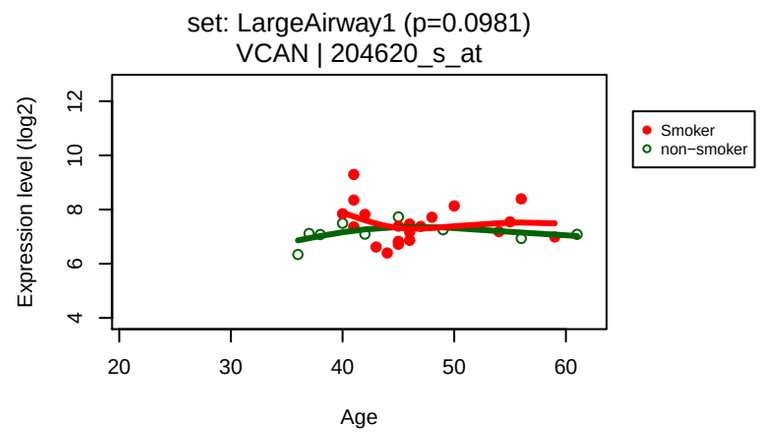

Gene expression of smokers and nonsmokers
gene: VCAN | 204620_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1812
p=0.0967
p=0.3518
p=0.0981
p=0.2114
p=0.0077
p=0.0512
p=0.0251
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

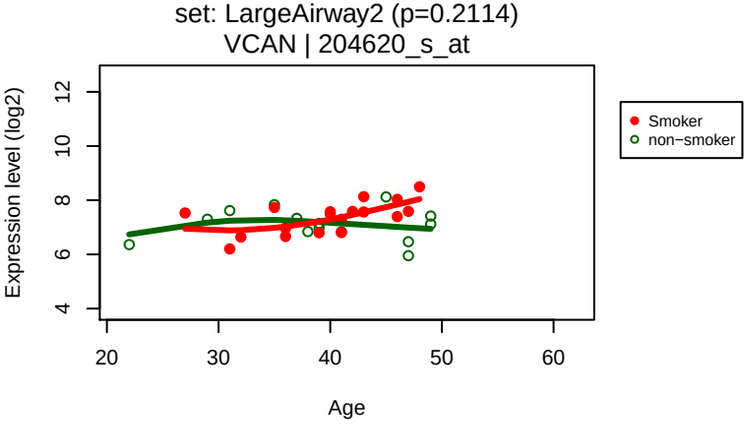

Gene expression ratio (smokers/nonsmokers)
gene: VCAN | 204620_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1812
p=0.0967
p=0.3518
p=0.0981
p=0.2114
p=0.0077
p=0.0512
p=0.0251
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

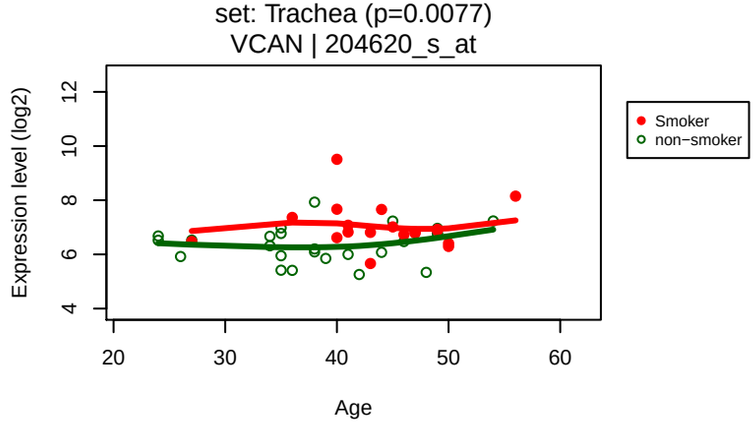

Gene expression ratio in Males
gene: VCAN | 204620_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2145
p=0.5472
p=0.0121
p=0.0327
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: VCAN | 204620_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2184
p=0.2732
p=0.4112
p=0.4533
SmallAirway2
LargeAirway2
Trachea
Macrophage2

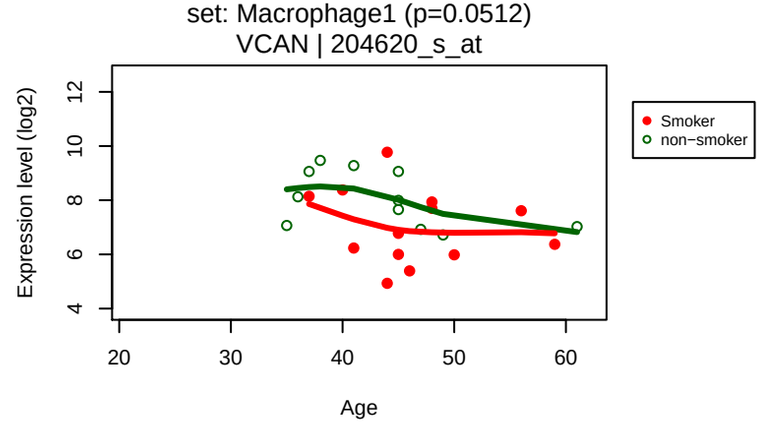

set: SmallAirway1 (p=0.1812)
VCAN | 204620_s_at
Expression level (log2)
Age
Smoker
non−smoker

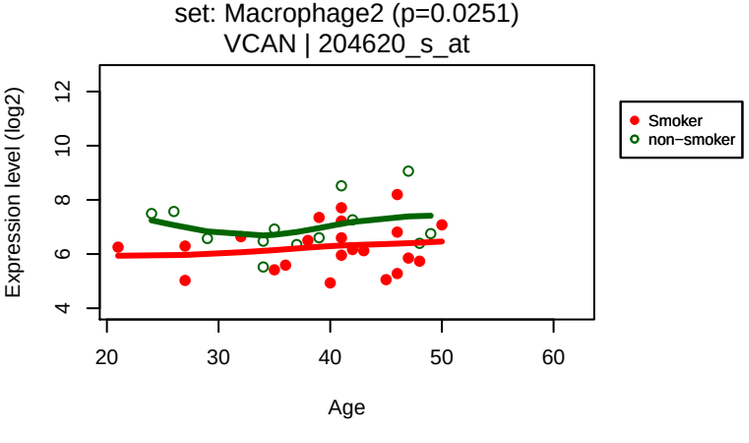

set: SmallAirway2 (p=0.0967)
VCAN | 204620_s_at
Expression level (log2)
Age
Smoker
non−smoker

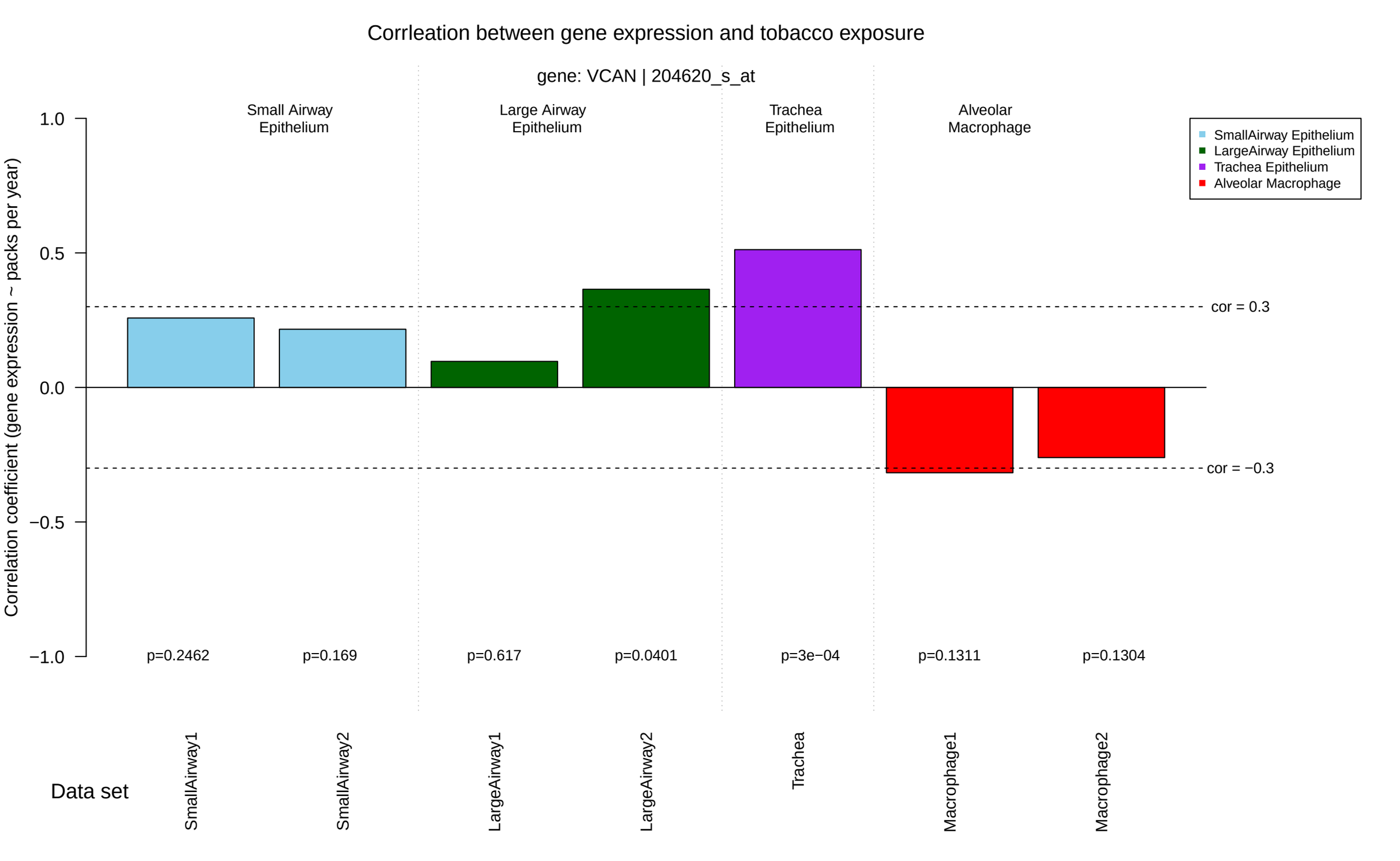

set: LargeAirway1 (p=0.0981)
VCAN | 204620_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2114)
VCAN | 204620_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0077)
VCAN | 204620_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0512)
VCAN | 204620_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0251)
VCAN | 204620_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: VCAN | 204620_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2462
p=0.169
p=0.617
p=0.0401
p=3e−04
p=0.1311
p=0.1304
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2